Risk Management: Assignment 5
Financial Engineering

Extended Vasicek model Monte Carlo simulation for CCR estimation

Instructions

- **Delivery:** Friday 18:00 the 15th of May to financial.engineering.polimi@gmail.com, with subject "RM: Assignment 5, Group #";
- Deliver
  a. pdf document with the results. List the errors you've found.
  b. the code written in Python (one notebook);
- Fill the gaps where appropriate and correct the errors in the code (in case of incoherences between this document and the code, follow this document);
- Comment the code and use self-explanatory variable names.
- Use the market data and the interest-rate curve bootstrapped in Assignment 0.

Case Study

It is February 15th 2008 and the IR-derivative desk of Polimi Bank has just entered a position in a vanilla 10y payer IRS. The floating rate is the 3-month Euribor (Q/Q, ACT/360 day count convention with a modified-following adjustment rule); the fixed rate has a yearly 30/360-European day count convention mod-foll. The risk management department, to evaluate the counterparty credit risk, decides to build a Monte Carlo with 250 000 simulations of the interest rate dynamics according to the Hull-White extended Vasicek interest rate model (see the code for the parameters).

Questions

1. Prove that for every simulation date s you can compute the discount factor curve as:

   $B(s,\tau) = A(s,\tau)e^{-x_s C(s,\tau)}$, where
   $A(s,\tau) = \frac{B(t_0,\tau)}{B(t_0,s)}e^{-\frac{1}{2}\int_{t_0}^{s}(\sigma(u,\tau)^2-\sigma(u,s)^2)du}$ and
   $C(s,\tau) = \frac{1-e^{a(\tau-s)}}{a}$

   Complete the code to perform the simulation of the discount factors over 40 equally spaced simulation dates using 250 000 scenarios. Parameters *a* and *sigma* defined in the code.
2. Compute the swap MtM at each simulation date and for each scenario
3. Derive numerically the expected positive exposure (EPE) profile (40 equally spaced future simulation dates);
4. Derive the 95% potential future exposure (PFE) profile (40 equally spaced future simulation dates);
5. From the PFE profile, derive the Peak-PFE;
6. Repeat the analysis above (2-3-4-5) assuming that Polimi Bank and counterparty (a-b-c are 3 independent scenarios)
   a. Agree to define a mandatory 3.5y break clause
   b. Agree to activate a netting agreement for the 10y swap payer with another swap that Polimi Bank and counterparty have traded, details in the code (compare the relevant exposures of the full portfolio, the sum of the relevant exposures of the 2 standalone trade and the single trades exposures)
   c. Agree to enter into a CSA agreement, with annual margin calls, such that the counterparty credit risk is significantly reduced at each collateral posting date. Choose the collateral parameters such that Polimi Bank is never required to post collateral,

while the counterparty covers fully its exposures, no IA. Suppose collateral is posted and adjusted immediately after a margin call and remunerated at the risk-free rate (use the shortest available zero rate as proxy)

7. Optional
   a. Compute an approximate CVA(t_0) estimation for each of the previous scenarios, under the assumption of LGD = 60%, exposures and default are independent, Polimi Bank cannot default, intensity of default of counterparty lambda=200bps flat.
   b. For the base case (no break, no netting, no CSA) try to compute 3 different kinds of sensitivities of the CVA (which are the main market risk factors/market calibrated parameters that affect CVA?)

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